Experiment 12: Project RESCUE-SWARM- Cooperative Multi-Agent Reinforcement Learning

Aim

To develop a multi-agent reinforcement learning system in which rescue agents learn to coordinate their actions in a disaster environment.

Exercises:

1. **Environment Design**: Create a grid-based environment with multiple rescue agents, victims, obstacles, and hazardous areas.
2. **States, Actions, and Rewards:** Define the states and possible actions of each agent. Assign rewards for successful rescues and penalties for collisions or unnecessary movements.
3. **Independent Q-Learning:** Implement Q-learning for individual agents and observe how they learn to complete rescue tasks independently.
4. **Cooperative Learning:** Extend the system to allow agents to coordinate their actions through shared information or common rewards. Compare the results with independent learning.
5. **Visualization:** Display agent movements and rescue progress. Plot learning curves showing how cumulative rewards change during training.
6. **Performance Evaluation:** Compare independent and cooperative approaches based on successful rescues, completion time, cumulative rewards, and collisions.